Zeal Education Society's

ZEAL COLLEGE OF ENGINEERING & RESEARCH, PUNE – 41

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

*NAAC Accredited with **A+** Grade / ISO 21001:2018*

Total No. of Questions: 08 | Total No. of Pages: 03

ZPRN: | Exam Seat No.:

Time: 02.00 Hour | Max. Marks: 60

[Paper Code - ADBS204]
F.Y. B.Tech. (Semester - II) / BTECH_ME_SEM II (End Term Examination)
Engineering Physics *(ITBS204/COBS204/ADBS204/AMBS204)*
(IT/COMPUTER/AIDS/AIML: 2024 Pattern)

General Instructions:

1. Figures to the right indicate full marks.
2. Draw neat labeled diagram wherever necessary
3. Attempt either **Q.1**OR **Q.2, Q.3**OR **Q.4, Q.5**OR **Q.6, Q.7**OR **Q.8**
4. Use h =6.63x10^{-34}J-S, m_e=9.1x10^{-31} Kg, e=1.6x10^{-19} C

Question No.		Question	Marks	CO	Blooms Level
Q.1	a)	State de- Broglie's hypothesis. Show that de- Broglie wavelength of an electron is inversely proportional to square root of applied accelerated Potential difference (V).	6	CO3	2
	b)	Explain the properties of matter waves. (Any five)	5	CO3	2
	c)	An electron has speed of 5.5x10^4 m/s with an accuracy of 0.003%. Calculate the uncertainty with which we can locate the position of electron.	4	CO3	3
OR					
Q2	a)	Explain principle, construction and working of Scanning Tunneling Microscope (STM) with the help of neat labeled diagram.	6	CO3	2
	b)	Obtain the wavelength of matter wave in terms of Kinetic Energy of moving particle.	5	CO3	2
	c)	Find the wavelength of electron which has velocity 18% of photons.	4	CO3	3
Q3	a)	What is Hall Effect? Derive the expression for Hall voltage and Hall coefficient and state its applications also.	6	CO4	2

	b)	Define Fermi energy level in semiconductor. Draw the diagram of P-type ad N-type semiconductors to show position of Fermi level.	5	CO4	2
	c)	A silver ribbon whose depth 2.3 cm and width is 0.4 cm. if the ribbon carries a current of 100 A from left to right and this ribbon is placed in transverse magnetic field of 1.0 T. Calculate the Hall voltage between the edges of ribbon. (Given Density of charge carries is $2.7 \times 10^{28}/m^3$)	4	CO4	3
OR					
Q4	a)	What is Solar Cell? Explain construction and working of solar cell with help of energy band diagram.	6	CO4	2
	b)	Draw IV characteristics of Solar cell and define the following terms: (i) Short circuit current (I_{sc}) (ii) Open circuit voltage (V_{oc}) (iii) Fill factor (iv) Efficiency of solar cell.	5	CO4	1
	c)	Calculate the mobility of charge carriers in doped silicon of conductivity 14.45 mho/m and the Hall coefficient is 1.57X 10^{-4} m^3/c.	4	CO4	3
Q5	a)	Differentiate between destructive testing and Non-destructive testing. (any six point)	6	CO5	2
	b)	Define compressibility and explain piezoelectric effect and inverse of piezoelectric effect with diagram.	5	CO5	1
	c)	An ultrasonic pulse of frequency 1.8 MHz is heard after 78 microseconds through the copper block. If velocity of ultrasonic in copper is 5800 m/s then find the thickness of copper block and also wavelength of ultrasonic waves.	4	CO5	3
OR					
Q6	a)	Explain Principal, Construction and Working of Ultrasonic Flaw Detection Technique for NDT with help of neat labeled diagram.	6	CO5	2

	b)	Define Ultrasonic waves on the basis of frequency and write its properties.(any 4)	5	CO5	1
	c)	Ultrasonic waves of frequency 4.5MHz and of wavelength 7.24×10^{-4}m is sent through a liquid of density 1000 $\frac{kg}{m^3}$. Calculate the velocity of ultrasonic waves in a liquid and compressibility of liquid.	4	CO5	3
Q7	a)	Differentiate between Type-I and Type-II superconductors.	6	CO6	2
	b)	Explain electrical property of nano particles.	5	CO6	2
	c)	The critical field of Niobium at 9 K is 2.5×10^5 A/m and at 0 K is 5.3×10^5. Calculate the critical temperature of the Niobium.	4	CO6	3
OR					
Q8	a)	State Meissner's Effect. Show that superconductor behaves like a perfectly diamagnetic material below critical temperature.	6	CO6	2
	b)	Explain mechanical property of nano particles.	5	CO6	2
	c)	A superconductor has critical temperature $8.5K$ and magnetic fields at 0 K is4.58×10^5 A/m. What is the critical magnetic field at 6 K.	4	CO6	3

*** Best of Luck ***